PHYS 210 Electricity and Magnetism

Celebration of Knowledge!

Practice Midterm H - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

Choose 3 of the following 4 problems. Complete every part of each chosen problem.

Write all rough calculations and final answers on separate paper. Show your working.

Original source numbering is retained inside each question. Answer the entire cropped question.

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1. [____] Homework 5 - Question 23.7

23.7 •• Two protons, starting several meters apart, are aimed directly at each other with speeds of 2.00×10^5 m/s, measured relative to the earth. Find the maximum electric force that these protons will exert on each other.

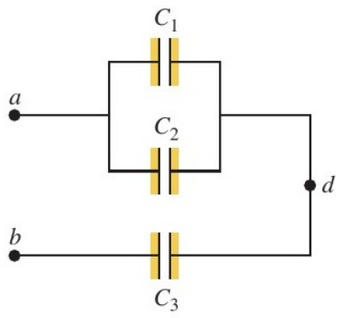
2. [____] Tutorial 4 - Question 6

6. Charge of uniform density (40 pC/m^2) is distributed on a spherical surface (radius = 1.0 cm), and a second concentric spherical surface (radius = 3.0 cm) carries a uniform charge density of 60 pC/m^2. What is the magnitude of the electric field at a point 4.0 cm from the center of the two surfaces?
   a. 3.8 N/C
   b. 4.1 N/C
   c. 3.5 N/C
   d. 3.2 N/C
   e. 0.28 N/C

3. [____] Textbook 15e - Question 24.55

24.55 •• In Fig. E24.20, $C_1 = 3.00\ \mu$F and $V_{ab} = 150$ V. The charge on capacitor C_1 is 150 μC and the charge on C_3 is 450 μC. What are the values of the capacitances of C_2 and C_3?

Figure **E24.20**



4. [____] Textbook 15e - Question VP24.11.1

VP24.11.1 The two plates of a parallel-plate capacitor each have area 0.460 m^2, are 3.00 mm apart, and initially have vacuum between them. A power supply is attached to the capacitor, charging it to 4.00 kV, and is then disconnected. A dielectric sheet is then inserted that fills the space between the plates. The potential difference between the plates decreases to 2.50 kV, and the charge on each plate remains constant. Find (a) the original capacitance, (b) the charge on the capacitor plates, (c) the dielectric constant of the dielectric, (d) the new capacitance after you insert the dielectric, and (e) the magnitude of the induced charge on the faces of the dielectric sheet.